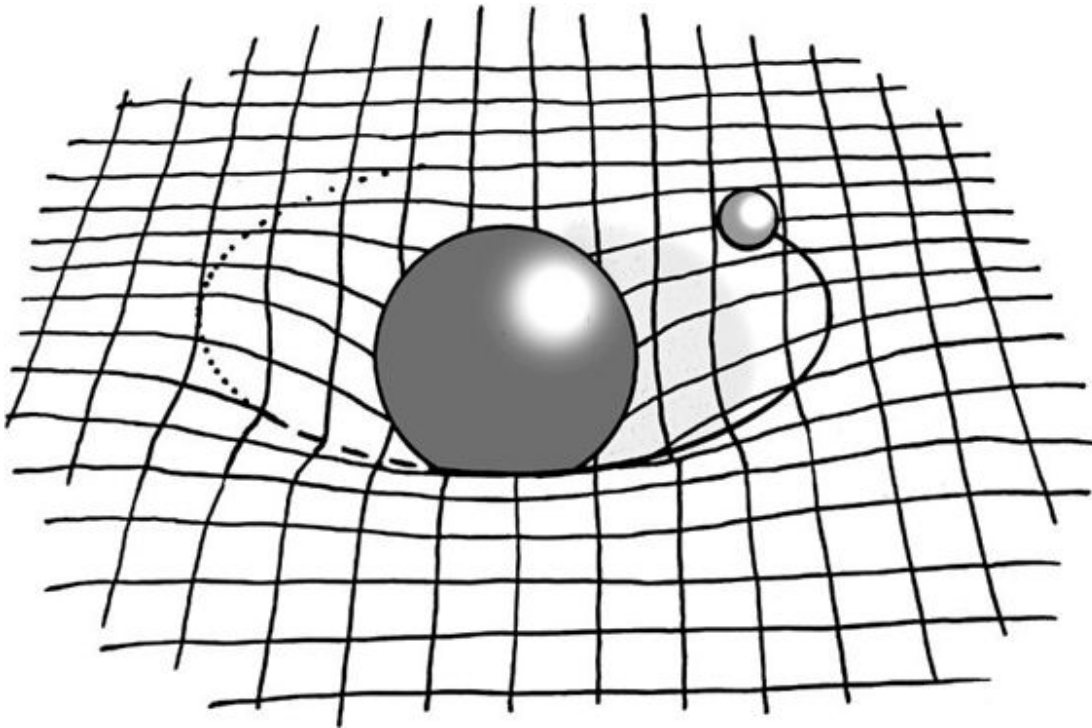


Paradigm shifts



Newton's ideas of gravity served as a framework for more than 200 years, but were overturned by Einstein's general theory of relativity, which is almost universally accepted by modern science.

The philosophy of science developed in the 20th century as an important branch in its own right, especially after Karl Popper's groundbreaking work. As well as examining the philosophical and logical basis of scientific methodology, philosophers turned their attention to the nature of scientific progress. It had been assumed that advances in science were a process of continuous evolution, until in 1962 Thomas Kuhn introduced the notion of paradigm shifts. In his analysis of the history of science, he suggested that periods of 'normal' science are interrupted by periods of 'crisis'. During the periods of normality, scientists work within an agreed framework, or paradigm, and any anomalies in their work are overlooked or dismissed. If, however, the anomalies become significant, this provokes a crisis, in which they have to be accounted for by new theories and a shift to a new paradigm. Normal science now resumes under the new framework, until the next crisis. Historical examples of such paradigm shifts include the revolutionary theories of Copernicus, Newton and Einstein.